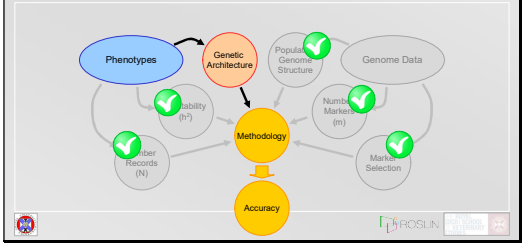
… Lecture 10 …
Mainly MCMC Models

Part 1:
The Bayesian Alphabet – up to C!

In this section, we are going to look at methods that allow more complex models for explaining the genetic variance than the ridge regressions underlying Van Raden models 1 and 2.

Factors for Accuracy



The genetic architecture of a trait is a broad term encompassing the number of loci contributing to its genetic variance and the distribution of its effects. This will include dominance and epistasis effects, and their contribution to the total genetic variance.

Van Raden Model 1

- Modelled the variance of allelic substitution
  - » assumed MVN(0, σ_b^2I)
  - » E[variance contributed | LE, HWE] = $2p_j(1-p_j)\,\sigma_b^2$

'given', i.e. 'conditional on'

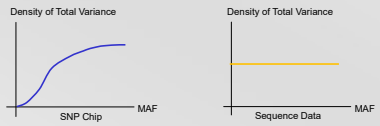
  - » locus of intermediate frequency expected to contribute more variance than one of low MAF

Return to Van Raden Model 1: it puts a variance on the allelic substitution effects (α) the same variance for all loci (σ_b^2). Therefore, the variance contributed by a locus is expected to be related to its MAF as the (genic) variance explained by a locus is $2p(1-p)\alpha^2$.

The contributions to the variance explained by the covariates will therefore depend on the MAF of the loci being used as regression covariates. If these covariate loci are from a SNP chip then there is a bias in all chips towards the most common alleles in a population, if only because they are the most easily discovered and verified. Low MAF are often included on SNP chips but are under represented.

Van Raden Model 1

- Modelled the variance of allelic substitution
  - » E[variance contributed | LE, HWE] = $2p_j(1-p_j)\,\sigma_b^2$
  - » where does variance of trait reside in model?

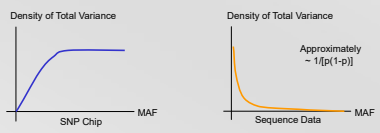


Theory and empirical data suggest that in the genome sequence the distribution of variants over MAF is strongly 'reversed-J' in shape with many more low MAF than high MAF. In fact, theory tells us that the distribution will not be far away from being proportional to 1/[p(1-p)].

This implies that with covariate loci from the sequence, the prior of the Van Raden Model 1is that the fraction of genetic variance contributed by <u>all</u> variants in an interval δp will be the same irrespective of p. With covariate loci from a SNP chip, the Van Raden 1 prior is that the variance explained will be strongly shifted to the covariate loci of high MAF, away from the low MAF. *Remember this is the variance explained in the model, and if we are using SNP chips we are using the idea of markers, and it does not strongly imply what the underlying model of causal variants may be. In contrast when using sequence data, the causal variants are in genotype data.* General <u>hypotheses</u>, e.g. 'rare alleles of large effect' are statements that are conditional on the variant being causative … '*if* I have a variant of large effect *then* it is likely to be rare'.

Van Raden Model 2

- Variance per locus was constant
  - » so variance in model follows SNP distribution



In Van Raden Model 2, the prior was that the variance contributed per locus used as a covariate was constant. Consequently, in Van Raden Model 2 the prior implies that with sequence data as the set of covariates then the great majority of genetic variance would be expected to reside with low MAF. When the covariate loci are from the SNP chip the majority of the genetic variance is associated with covariate loci of high MAF.

More Models

- What distribution of variance per locus?
- Why is every locus expected to contribute?

However underlying both the Van Raden models, and in the corresponding $\mathbf{G}_1$ and $\mathbf{G}_2$, are assumptions of Normal distributions of substitution effects. Is this reasonable?

Bayes A

- Assume each SNP locus has a b_j
  - » **y** ~ **1**m + … + **Xb** + **e** as before
  - » b_j now come from scaled t-distribution
    - » not from $N(0,\sigma_b^2)$, i.e. scaled Normal
  - » t-distribution has 'heavier tails'
    - » 'shrinks small effects more than Normal'
    - » 'shrinks large effects less than Normal'

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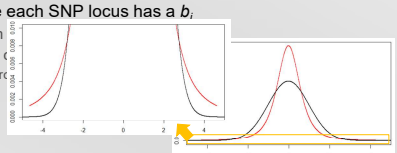
Why not assume the substitution effects (the *b*'s in the slide) come from a Student's t-distribution? This has heaver tails than a Normal distribution. It gives greater prior credibility to the idea of some large effects, so shrinks small effects more than Normal, and large effects less than Normal.

This model was introduced in the original Meuwissen et al. (2001) publication, where it was called 'Bayes A'.

Bayes A

- Assume each SNP locus has a b_j
  - » **y** ~ **1**m
  - » b_j now
    - » not fr
  - » t-distribution has 'heavier tails'
    - » 'shrinks small effects more than Normal'
    - » 'shrinks large effects less than Normal'



Kurtosis: platykurtic (light-tailed) or leptokurtic (heavy-tailed) c.f. Normal

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Student's t-distribution is termed leptokurtic. Leptokurtic distributions have heaver tails than a Normal distribution with the same variance.

Bayes B

- Assume some SNP have no effect
  - » say fraction π, pre-determined
- Other (1-π) have effects
  - » distributed as Bayes A with scaled t-distribution

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The Meuwissen et al. (2001) publication also introduced a model called 'Bayes B'. Here it is assumed that a fraction π of the loci have no effect at all, whereas the other (1-π) are distributed as in 'Bayes A'. In the publication π was a pre-determined constant. After Meuwissen et al. (2001) it became fashionable to invent variations in this theme and extend 'the Bayesian alphabet'.

Bayes Cπ

- Assume some SNP have no effect
  - » say fraction π
  - » π ~ Uniform[0,1]
    - » i.e. estimate π
- Other (1-π) have effects
  - » distributed as Normal
  - » i.e. NOT t-distribution

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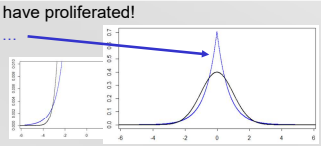
For example, 'Bayes C', which is 'Bayes B' but where the distribution of the non-zero effects is assumed to be Normal.

Then there is 'Bayes Cπ' in which π is now estimated, e.g. with a uniform prior on [0,1].

Bayesian Alphabet

- Such models have proliferated!
  - » e.g. Laplace …



- Common theme
  - » reduce shrinkage of apparently large effects
  - » QTL hypothesis

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Minor variations on these ideas have proliferated using different letters of the alphabet: using more than two buckets of loci, e.g. Bayes R; or using different distributions for the effects, typically with heavy tails, e.g. a Laplace distribution which is a 2-sided symmetric exponential distribution.

Bayes Alphabet

- Now have priors on
  - » fractions of marker loci having an effect
  - » distributions of marker effects or variance
  - » forms of subset selection
    - » identifying subsets with small or large effects
- The priors on distributions result in shrinkage
  - » priors are random distributions on marker effects
  - » functional form of shrinkage depends on the prior

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So, there are many aspects that one can put priors on: fractions with non-zero effects, distributions of non-zero effects, buckets with effects of different sizes each with their distribution.

Bayes Alphabet

- Models are simple to write down - but complex to solve!
- Use Monte Carlo Markov Chains
  - » MCMC
  - » can be applied if one can write down the conditional distributions

… Lecture 10 …
Mainly MCMC Models

Part 2:
'The Cat at Christmas' or 'It's a Cat's Life'

Markov Chain Example

- Suppose I have a cat
  - » cat is either indoors or outdoors all day & night
  - » cat may move indoors/outdoors only when I leave for work and remains until next morning
  - » what happens depends on where I find the cat
  - » the transitions are constant over time

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

Markov Chain Example

- I have defined the conditional distributions
  - » P[Inside at t+1 | Inside at t] = 0.6
  - » P[Outside at t+1 | Inside at t] = 0.4
  - » can define conditional distribution if 'Outside at t'
  - » i.e. given any current state I can define the full conditional distribution for moving to another

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

It's A Cat's Life

- What is the probability that on Christmas Day the cat will be inside?
  - » I can solve by some fancy maths!

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

It's A Cat's Life

- What probability that on Christmas Day the cat will be inside?
  - » I can also solve by sampling random numbers
  - » Monte Carlo Markov Chain (MCMC)

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

These models are simple to write down but very complex to solve. These require Monte Carlo Marko Chains (MCMC) for solutions which can be applied if one can write down the conditional distributions for each effect.

It's A Cat's Life

- Start by assuming an S(0), say 'In'
- Draw random number X in U[0,1]
  - » if 'In' & X ≤ 0.6, stay 'In', else 'Out'
  - » if 'Out' & X ≤ 0.2, go 'In', else stay 'Out'
- If done, say, 1000 times, stop, else go to 2

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

It's A Cat's Life

- Initial values will depend on starting point
- After a while influence of S(0) decays, burn-in
- Then sequence is sample from an equilibrium
- Look at how frequent 'In' occurs in sample
- This is the probability of being 'In' on 25/12

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	0.8	0.4
	Cat is Inside	0.2	0.6

It's A Cat's Life

	Burn In	0	10	20	30	40	50	60	70	80	90
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											

		From	
		Cat is Outside	Cat is Inside
To	Cat is Outside	1, 2, 3, 4, 5, 6, 7, 8	1, 2, 3, 4
	Cat is Inside	9, 10	5, 6, 7, 8, 9, 10

It's A Cat's Life

- Our answer is 0.33
- Analytical answer is 1/3!
- MCMC with 100,000 estimate is 0.3342
  - » s.e. ~ 0.0039

… Lecture 10 …
Mainly MCMC Models

Part 3:
MCMC & the amazing Ergodic Theorem!

Monte Carlo Markov Chains

- Have n states that describe the problem
  - » 2 in cat example
- Have complete set of conditional distributions
  - » what are the probabilities of the next state given where I am now

FROM

TO		Cat is Outside	Cat is Inside
	Cat is Outside	1 to 8	1 to 4
	Cat is Inside	9 to 10	5 to 10

- Can obtain the distribution of solution

The cat is very simple problem with just two states 'Inside' or 'Outside'. It is perhaps very easy to see how this can be extended to 3, 4 … *n* states for some *n* which is possibly large.

Monte Carlo Markov Chains

- Can go from 2 to n states S_1, S_2 … S_n
  - » if we can define $P[S(t+1)=S_i \mid S(t)=S_j] = c_{ij}$
  - » for all i and j, c_{ij} remains constant over time
  - » e.g.

FROM

TO

	Cat is Absent	Cat is Outside	Cat is Inside
Cat is Absent	0	0.05	0.10
Cat is Outside	1	0.35	0.50
Cat is Inside	0	0.60	0.40

For example, here is a 3-state problem, the cat is now absent sometimes in the mornings!

Monte Carlo Markov Chains

- Can go from 2 to n states S_1, S_2 … S_n
  - » if we can define $P[S(t+1)=S_i \mid S(t)=S_j] = c_{ij}$
  - » for all i and j, c_{ij} remains constant over time
  - » the c_{ij} are conditional probabilities … 'probability of moving to i given now in j'
  - » i.e. can write down conditional distributions for moving from one state to another

What is required is to write down the set of conditional probabilities of moving from one state to another.

Monte Carlo Markov Chains

- Can go from 2 to n states S_1, S_2 … S_n
  - » if we can define $P[S(t+1)=S_i \mid S(t)=S_j] = c_{ij}$
  - » for all i and j, c_{ij} remains constant over time
  - » the c_{ij} are conditional probabilities … 'probability of moving to i given now in j'
  - » i.e. we can write down the conditional distributions for moving from one state to another
- Can use Monte Carlo Markov Chains
  - » can find probabilities of states from 'stationary' distribution

Then carry through sampling procedures until the stationary distribution over the states is reached.

Continuous Variables

- Observe data **y**
- Model **y** with parameters **u**
  - » **u** will have prior distributions, $\theta(\mathbf{u})$
- Can consider posterior distribution of **u** given (conditional on) **y** and priors $\theta(\mathbf{u})$
- Our states become all possible values of **u**
- Infinite number of states if **u** continuous
- Solution is probability distribution (density) over all the states

This can be generalised to a set of states whose values are continuous variables **u**, e.g. breeding values, but with some known data **y**, e.g. phenotypes.
These may be assumed to have distributions from which they have been sampled, e.g. a random effect, with a 'prior' distribution $\theta(\mathbf{u})$, e.g. $MVN(0,V_A\mathbf{A})$. The use of the priors indicate we are entering **Bayesian statistics** – more in Tutorial 10.
What is wanted is the distribution of **u** *conditional on* observing the data and our assumed priors. This is called a 'posterior' distribution.
Now there is an infinite number of states, with the states corresponding to the parameters!
Amazingly (to me) there is the Ergodic Theorem – note the '(Amazing)' in the slide is my addition! This says that if the conditional distribution of each parameter in **u** can be written down assuming all the others are known and given the data **y** and the priors, then the posterior distribution can be obtained by sampling schemes just like 'The Cat at Christmas'.

The (Amazing) Ergodic Theorem

If I have data **y** … continuous or categorical

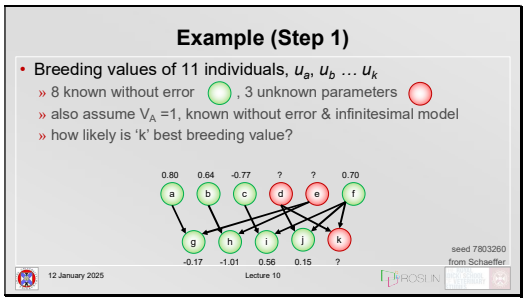
1. List parameters $\mathbf{u} = u_1, u_2 \ldots u_n$ with data **y**
2. Write down all conditional distributions
   i.e. $u_i \mid u_1, u_2 \ldots u_{i-1}, \quad u_{i+1}, u_n, \mathbf{y}$ for all i
3. Find some starting values, $\mathbf{u}^0$, say $u_1^0, u_2^0 \ldots u_n^0$
4. Sample from conditional distributions e.g. if n=3
   $\mathbf{u}^1 = (u_1^1 \mid u_2^0, u_3^0, \mathbf{y}), (u_2^1 \mid u_1^1, u_3^0, \mathbf{y}), (u_3^1 \mid u_1^1, u_2^1, \mathbf{y})$

The (Amazing) Ergodic Theorem

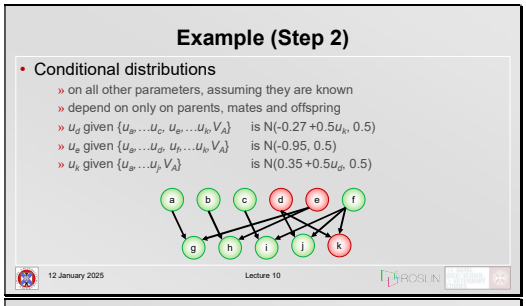
If I have data **y** … continuous or categorical

1. List parameters $\mathbf{u} = u_1, u_2 \ldots u_n$ with data **y**
2. Write down all conditional distributions
   i.e. $u_i \mid u_1, u_2 \ldots u_{i-1}, u_{i+1}, u_n, \mathbf{y}$ for all i
3. Find some starting values, $\mathbf{u}^0$, say $u_1^0, u_2^0 \ldots u_n^0$
4. Sample from conditional distributions e.g. if n=3
   $\mathbf{u}^1 = (u_1^1 \mid u_2^0, u_3^0, \mathbf{y}), (u_2^1 \mid u_1^1, u_3^0, \mathbf{y}), (u_3^1 \mid u_1^1, u_2^1, \mathbf{y})$
5. Repeat Step 4, using $\mathbf{u}^t$ to get $\mathbf{u}^{t+1}$ … to large t

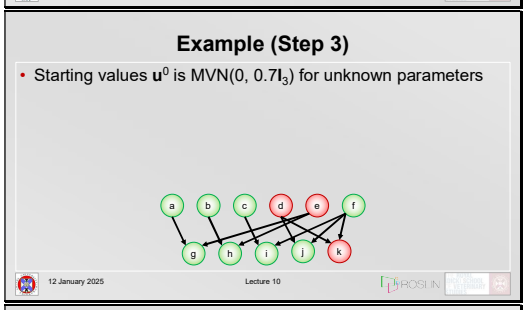
One continues sampling until one has a very large sample from the stationary distribution over all possible values of **u**.



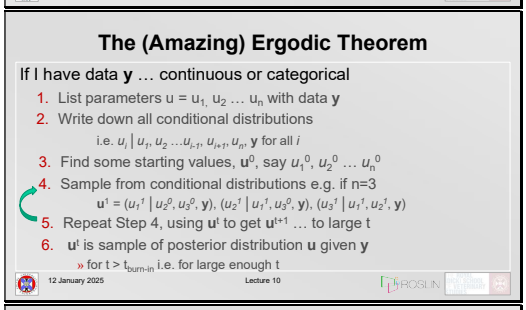
Consider a Bayesian approach to the following problem. Breeding values of 11 individuals, with 8 known without error (our data **y**) and also assume that the genetic variance V_A is also known to be 1 without error, i.e. 'the prior for V_A has probability 1 for V_A=1, and probability 0 for all other values'.
So I have only 3 states, the breeding values of *d*, *e* and *k*, but all are continuous variables. I can write down the conditional distributions for each of these 3 states, assuming all the others are known, i.e. **all the remaining 10**. (It is **not** necessary for QGM to know how these were obtained.)
u_e ~ N(-0.95,0.5), where the -0.95 comes from knowing *a*, *b*, *g*, *i*, and knowing any of the others does not add information on what the breeding value of *e* is.
u_d ~ N(-0.27+0.5u_k,0.5), where the -0.27 comes from knowing *j* and *f* and it will also depend on whatever the state of *k* is when the value of u_d is sampled.
u_k ~ N(0.35+0.5u_d,0.5), where the 0.35 comes from knowing *d* and *f* and it will also depend on whatever the state of *d* is when the value of u_k is sampled.



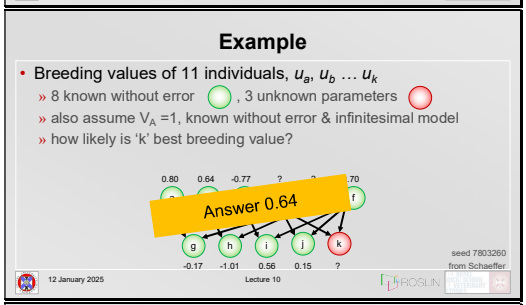
Get some starting values for the 3 unknowns.



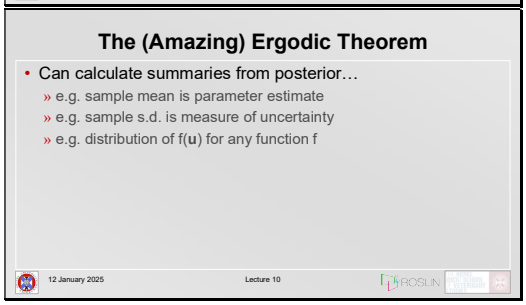
Apply the theory …



The probability that *k* has the highest breeding value among the 3 unknown individuals is 0.64.



What we get from MCMC is a large sample from posterior distributions. Providing the sample is large enough then the distribution will be very close to the true posterior distribution.
e.g. the mean of the posterior distribution is typically used as the estimate, the s.d. of the posterior is a measure of uncertainty like the s.e.
A valuable property used above that the distribution of f(**u**) can also be obtained from the same large sample e.g. f() may be max(), as used above.



The (Amazing) Ergodic Theorem

- Can solve complex models by defining all conditional distributions $u_i \mid u_1, u_2 \ldots u_{i-1}, u_{i+1}, u_n, \mathbf{y}$
- Must ensure *stationarity* (c.f. convergence)
  - » posterior does not depend on starting values
  - » all regions of posterior distribution explored
    - » more parameters, more cycles
    - » 'mixing speed' slow for genotypes
- Time = # parameters x # cycles x 'unit cpu'
  - » cycle number is large … say multiples of 10,000!

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MCMC does have its issues. The samples have to be large to establish stationarity, i.e. it does not depend on the starting values, the whole of the posterior distribution is explored. So many cycles of sampling are required. This takes time as there are large numbers of random samples needed: # parameters x # cycles.

The (Amazing) Ergodic Theorem

- Sampling procedure used was 'Gibbs Sampling'

$$\mathbf{u}^{t+1} = (u_1^{t+1} \mid u_2^t, u_3^t, \mathbf{y}), (u_2^{t+1} \mid u_1^{t+1}, u_3^t, \mathbf{y}), (u_3^{t+1} \mid u_1^{t+1}, u_2^{t+1}, \mathbf{y})$$

  - » slow
  - » more complex procedures can speed mixing and stationarity
- Generally, need to sample from many different distributions
  - » e.g. priors on variances are 'inverse chi-squared'
  - » avoided in example by assuming V_A known, i.e. Prob(V_A=1) = 1
- Use of 'uninformative priors'
  - » how much have the priors influenced the results

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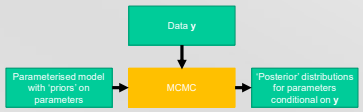
The sampling strategy illustrated is called 'Gibbs Sampling' (where the parameters are sampled cyclically one after another). This is often easy to do, but slow. There are other sampling strategies which are faster.
One needs to be able to sample from a wide range of distributions e.g. sampling variance parameters requires sampling from 'inverse-chi-squared' distributions.
Setting prior distributions in Bayesian statistics values is always an issue, as these will influence the outcomes. It is philosophically controversial which are touched upon in Tutorial 10. Generally, minimally informative priors are used.
In the above I have approached Bayesian statistics from their use in quantitative genetic models. A consequence of this is that the Bayesian ideas have been merged and coupled with the use of MCMC. Bayesian statistics is a philosophical approach to statistical inference that differs from the frequentist approach which has been adopted in this course until now. This is touched upon in Tutorial 10. Bayesian statistics does **not** require MCMC, but MCMC makes many Bayesian models tractable.

'Bayesian Methods' & MCMC



» Bayesian Methods do not require MCMC
» MCMC makes complex models feasible

Thomas Bayes studied Logic & Theology, University of Edinburgh 1719-1722!

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MCMCglmm

- R package
  - » by Jarod Hadfield
  - » use for Blue Tit data in Tutorial 10!

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… Lecture 10 …
Mainly MCMC Models

Part 4:
Back to the Bayesian Alphabet

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Bayesian Alphabet

- Bayes A, B, C, … can be solved using MCMC
  - » as we can write down all conditional distributions
    - » and sample from them
  - » computationally snail-paced c.f. GBLUP
  - » there are fast 'approximate' algorithms

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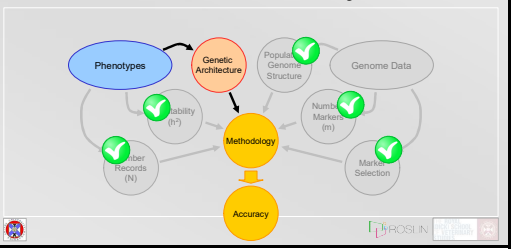
Factors for Accuracy



Daetwyler et al (2010) demonstrate that GBLUP is indifferent to the genetic architecture. Accuracy changes very little as the structure of the genetic variance changes.

Bayesian Methods

- Bayesian methods can put prior on numbers of contributing loci
  - » e.g. Bayes B
- When better than GBLUP?
  - » GBLUP treats as if *Me* 'loci'
    - » 'number of beans in bean bag'
  - » prior useful if # QTL less than *Me*
  - » best evaluation method depends on population genome
    - » i.e. 'number of beans in population's bean bag'

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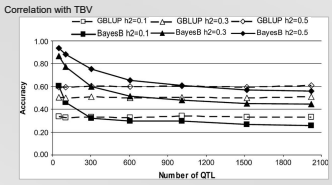
Putting priors on the number of loci as in Bayes B, and by inference other methods that rely upon partitioning loci into sets of different sizes, can perform much better when there are only small numbers of QTLs that explain almost all the variance, but only perform as well as GBLUP when the numbers of QTL are large. Therefore, GBLUP is generally conservative in accuracy and planning with GBLUP in mind is a safe approach.

Accuracy as f(#*QTN*)



From Daetwyler et al. 2010

QTNs in data, 'sequence-ready'; N_e =1000, L=10M, N=1000

12 January 2025 Lecture 10

GBLUP treats the genome as if there are M_e 'independent segments', where M_e characterises the species genome structure arising from the extent of LD. Therefore, what is a small number of QTL will depend on M_e: what is a small number for humans with large N_e and M_e, may not be small for dairy cattle with (relatively) small N_e and M_e. Methodological benefits in accuracy of Bayes B (and similar methods) will therefore depend on the genetic architecture of the traits in relation to the genome structure of the species (or population) in which the trait is measured.

This helps to understand the outcomes of some simulation studies. In early studies Bayes B was always much better than GBLUP and this was accepted as 'a methodological fact' until real data became available with sufficient density of SNP, primarily in dairy cattle. Then it appeared that GBLUP was as good – or perhaps marginally better! This can now be seen as reflecting the traits (yields of milk with many causative mutations) and the species genome (low N_e and low M_e). Early simulations were careless about the number of QTL and structure of the genome, and had too few QTL determining the trait.

Bayesian Alphabet

- Early simulations always had few QTN cf M_e
- Always found BayesB better!
- Practical cattle data had no benefit of BayesB
- Relative merit of method depends on species
  - » dairy cattle, low N_e, GBLUP as good
  - » medical, high N_e, BayesB-type more likely a benefit
  - » however need to compute!
- Changing distributions has only small effect

12 January 2025 Lecture 10

Finally, to summarise the potential impact of genomic models on breeding, which has been my main area of activity, I wish to revisit QG.

As a generalisation, the difficult problems with breeding are where selection is needed on sex-limited traits (only one sex expresses the trait), age-limited traits (the trait is expressed late in life and delays selective breeding), or destructive traits (the trait is only measured after death).

… Lecture 10 …
Mainly MCMC Models

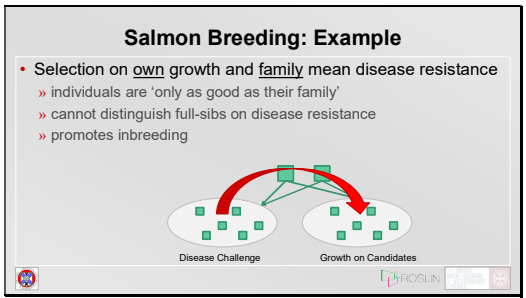
Part 5:
QG Revisited & the Impact of Genomics

12 January 2025 Lecture 10

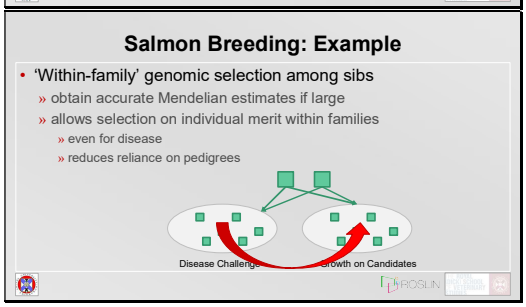
Salmon Breeding: Example

- Important objective is disease resistance
- Candidates must be free of disease
- Large full-sib families, each generation
  - » split into two populations
    - » one for breeding
    - » one for exposing to pathogens & recording mortality

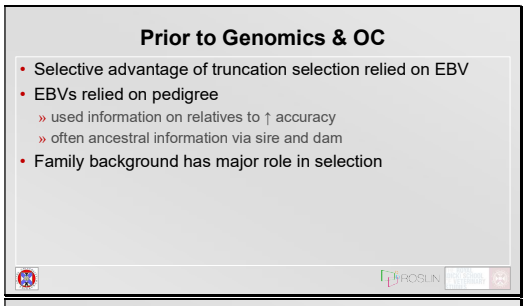
To illustrate the impact of genomics, consider an example of salmon breeding. Here an important objective is to improve disease resistance, but candidates must be free of disease, and so they are never exposed to it. So where is the information for breeding to come from? This problem can be overcome in salmon by utilising their very large full-sib families. The family is sub-divided into two parts: one is exposed to the pathogen, and the other is kept disease free and supply the candidates for selection.

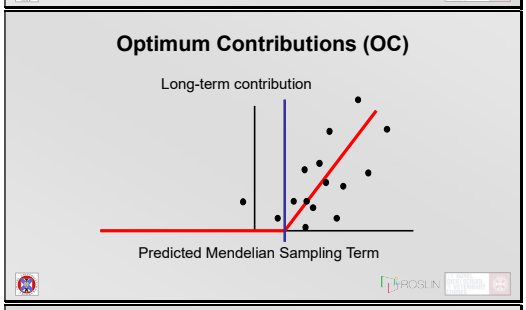


Whilst this provides valuable information, it poses a genetic challenge. The index of selection is based on the growth of the candidate, which can be measured directly on the candidate (and its relatives) and the family mean for disease resistance. Therefore, full-sibs cannot be distinguished on disease resistance, and this has two consequences: (i) it limits the accuracy, and hence progress, for disease resistance: (ii) it promotes inbreeding as all the full-sib candidates share precisely the same information on disease resistance.

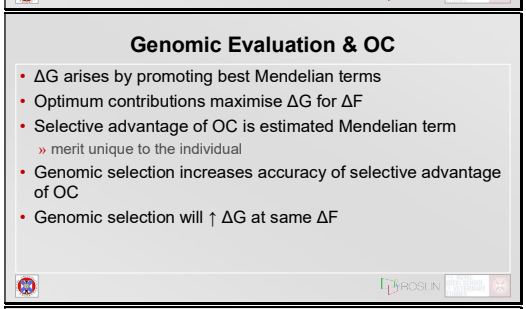


However, with genomics, the disease challenge full-sibs can be genotyped and good predictions can be obtained of the ranking within the family (it does not require very large numbers of full sibs to achieve this, perhaps a few tens, see a paper by Lillehammer et al. 2014). This ranking within the family is possible because the genomics is estimating the Mendelian sampling term of the individual as it is possible to observe which alleles were passed by the parent to the individual. This prediction can then be applied to the candidates after genotyping them. Consequently, the candidates now have predictions of its own Mendelian sampling term and they can now be distinguished from its full-sibs. This reduces the reliance on pedigrees.

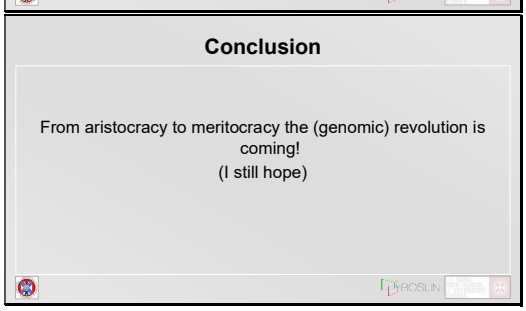




Remember Optimum Contributions (OC), which maximises gain for a given rate of inbreeding. In such a selection scheme the selective advantage was the estimated Mendelian sampling term (c.f. truncation selection where the entire EBV or TBV is the selective advantage).
(Note the genomics will increase accuracy of the sire and dam components too, but this is not the key point in the argument!)



Therefore, optimum contributions promotes those individuals with the best Mendelian sampling terms. The use of genomic data for evaluation in genomic selection (GS) increases the accuracy of predicting these Mendelian sampling terms, and consequently offers more progress at the same rate of inbreeding.



Traditional BLUP has relied on pedigrees and typically increases accuracy through information on collateral relatives, increasing the information on the sire and dam components. Success was often based on how good the sire and the dam of the candidate was considered to be – an aristocratic system. With genomics, it is now possible to assess what new merit a candidate is bringing to the population – a meritocratic system.

A true revolution … driven by genomics!